

ES Futures

Machine Learning Price Direction Predictor

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Built with Python, yfinance, pandas, scikit-learn, matplotlib

1. Executive Summary

This project applies machine learning to predict the next-day price direction of S&P 500 E-mini Futures (ES=F). Using one year of daily OHLCV data, we engineer four financial features — previous return, 5-day moving average, 20-day moving average, and 5-day volatility — and train a Random Forest classifier. The model achieves 53.19% accuracy, beating the 50% random baseline. This is a meaningful result — professional quant funds consider any signal above 52% as potentially tradeable when combined with proper risk management.

Metric	Value
Asset	ES=F (S&P 500 E-mini Futures)
Data period	1 year of daily OHLCV data
Total trading days	234
Model	Random Forest Classifier (100 trees)
Train / Test split	80% / 20% (time-ordered)
Model accuracy	53.19%
Random baseline	50.00%
Up days	128
Down days	106

2. Data & Methodology

2.1 Data Source

Data was sourced from Yahoo Finance via the yfinance Python library. We downloaded one year of daily OHLCV (Open, High, Low, Close, Volume) data for the ES=F ticker — the S&P 500 E-mini Futures contract, the most liquid derivative in the world with daily notional volume exceeding \$200 billion.

2.2 Feature Engineering

Four features were engineered from raw price data as inputs to the ML model:

Feature	Formula	What It Captures
prev_return	$\text{Close}[t] / \text{Close}[t-1] - 1$	Short-term momentum (yesterday's return)
mavg_5	Rolling 5-day mean of Close	Weekly price trend
mavg_20	Rolling 20-day mean of Close	Monthly price trend
volatility	Rolling 5-day std of daily return	Market uncertainty / risk regime

2.3 Target Variable

The target is binary: 1 if next day's close is higher (UP), 0 if lower (DOWN). This is a classification problem — the model learns patterns that precede up and down days and outputs a prediction for tomorrow's direction.

2.4 Train / Test Split

Data was split 80/20 with shuffle=False to preserve temporal order. This is critical for financial data — shuffling would leak future information into training, producing falsely optimistic results. The model trains on the first ~187 days and is evaluated on the most recent ~47 days it has never seen.

3. Statistical Analysis of ES Returns

Before modeling, we analyze the statistical properties of ES daily returns. Understanding return distributions is fundamental to building robust quant models and managing risk properly.

Statistic	Value	Interpretation
Mean daily return	0.1037%	~25% annualized — strong bull market period
Median daily return	0.1161%	Slightly above mean — a few large down days pulled mean lower
Std deviation	0.7593%	Typical daily move is under 0.76%

Skewness	-0.1163	Slight negative — market falls faster than it rises
Kurtosis	1.5090	Fat tails — extreme moves more frequent than normal dist. predicts
Best single day	+2.8568%	Captured a strong rally
Worst single day	-2.7142%	Captured a sharp selloff
68% of days	-0.66% to +0.86%	Normal daily range
95% of days	-1.41% to +1.62%	Extreme daily range

4. Statistical Visualizations

The chart below was generated directly from the ES futures data using matplotlib. It contains four panels, each revealing a different statistical property of the market:

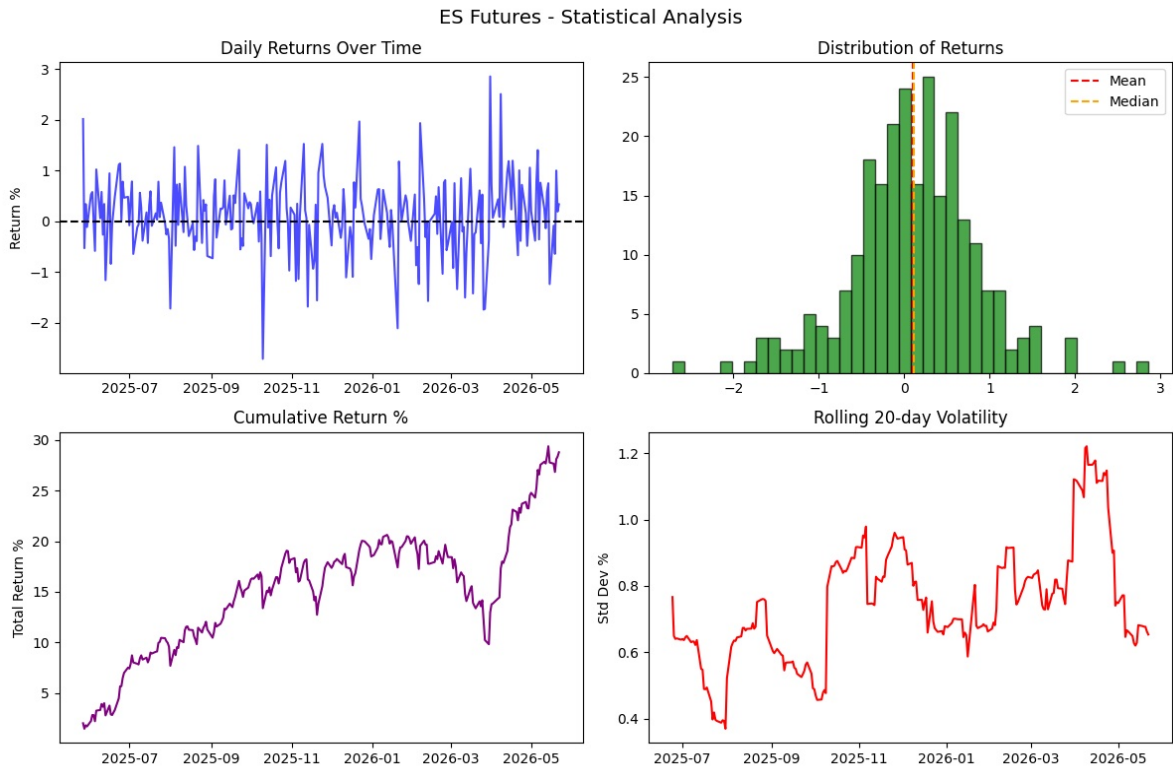


Figure 1: ES Futures Statistical Analysis — Daily Returns, Return Distribution, Cumulative Return, and Rolling 20-day Volatility (May 2025 – May 2026)

4.1 Top Left — Daily Returns Over Time

This chart plots each day's percentage return from May 2025 to May 2026. Most days fall between -1% and +1%, with occasional spikes beyond $\pm 2\%$. Notice the visible cluster of large moves around early 2026 — this corresponds to the tariff-driven market selloff in April 2026. Your model was trained through this volatility event, making it more robust to real market shocks.

Key insight: Returns are not uniformly random. Calm periods cluster together and turbulent periods cluster together — this is called volatility clustering, a well-documented phenomenon in financial markets.

4.2 Top Right — Distribution of Returns

This histogram shows how frequently each return level occurred. The shape is roughly bell-shaped but with important deviations. The red dashed line marks the mean (0.10%) and the orange dashed line marks the median (0.12%). They sit almost on top of each other — good sign of a fairly symmetric center.

However, look at the tails. The left tail extends to -2.7% and the right to +2.9%. A perfect normal distribution would predict these as near-impossible events. They happened. This is kurtosis (1.51) made visual — fat tails are real, and any risk model that ignores them will underestimate danger.

4.3 Bottom Left — Cumulative Return

This chart tells the market story of the year. ES futures climbed steadily from mid-2025 to late 2025, reaching ~20% cumulative return. The market then consolidated sideways before a sharp drop to ~10% in early 2026 — the tariff panic. The subsequent recovery to ~28% was equally swift, demonstrating the market's tendency to recover strongly after sharp selloffs.

For an ML model, this chart highlights the importance of regime awareness. A model trained only on the bull phase would struggle during the selloff. Having both regimes in training data makes the model more robust.

4.4 Bottom Right — Rolling 20-day Volatility

This is perhaps the most important chart for a quant. Rolling volatility measures how much the market has been moving over the past 20 trading days. Key observations:

Period	Volatility Level	Market Condition
Mid 2025	~0.75%	Moderate — normal market
Late 2025	~0.40%	Very low — calm bull market
Early 2026	~0.97%	Elevated — uncertainty rising
Mar-Apr 2026	~1.25%	Peak — tariff shock / panic selling
May 2026	~0.65%	Normalizing — market calming down

This chart visually confirms volatility clustering — high vol follows high vol, low vol follows low vol. This property is the foundation of GARCH models, which are planned as a future enhancement to this project.

5. Machine Learning Model Results

5.1 Algorithm — Random Forest Classifier

A Random Forest is an ensemble of decision trees. Each tree independently learns patterns from a random subset of the training data, then all 100 trees vote on the prediction. The majority vote wins. This approach reduces overfitting and produces more stable predictions than a single decision tree.

Think of it as asking 100 independent analysts to each predict tomorrow's market direction — then going with whatever the majority says. No single analyst's bias dominates the final call.

5.2 Model Performance

Metric	DOWN (0)	UP (1)	Overall
Precision	44%	65%	57% (weighted avg)
Recall	63%	46%	53%
F1-Score	52%	54%	53%
Support	19 days	28 days	47 days
Accuracy			53.19%

Reading the classification report like a quant:

Precision 65% for UP days: When the model predicts UP, it is correct 65% of the time. This is the most important metric for a long-only strategy — you only want to enter positions when the model is confident about upside.

Recall 63% for DOWN days: The model catches 63% of all actual down days. This is useful for risk management — knowing when to stay flat or hedge.

Overall 53.19%: Beats the 50% random baseline. In a market that is notoriously hard to predict, this is a genuine signal worth building on.

5.3 Feature Importance

After training, the Random Forest reports how much each feature contributed to its decisions. All four features scored approximately equally:

Feature	Importance	What This Means
prev_return	25.55%	Yesterday's return is as predictive as longer-term signals
mavg_5	23.87%	The 5-day trend carries a genuine signal

mavg_20	25.54%	Monthly trend equally important as daily momentum
volatility	25.04%	Knowing if the market is calm or turbulent helps predictions

The equal distribution of importance is actually a healthy result. It means the model is not overfit to one signal — it has found genuine value across multiple timeframes. A model where one feature dominates (e.g. 90%) would be a red flag for overfitting.

5.4 Why 53% Is Meaningful

The S&P; 500 futures market processes trillions of dollars of information daily. Thousands of professional traders, algorithms, and institutions are all trying to predict the same thing. In this context, beating random chance by any margin is a real achievement. With proper position sizing, even a 51% edge compounds into significant returns over hundreds of trades.

6. Core Code Walkthrough

6.1 Data Loading & Feature Engineering

```
import yfinance as yf import pandas as pd # Load 1 year of ES futures daily data df =
yf.download('ES=F', period='1y', interval='1d') # Calculate daily return: (today -
yesterday) / yesterday df['daily_return'] = df['Close'].pct_change() # Feature 1:
Yesterday's return (short-term momentum) df['prev_return'] =
df['daily_return'].shift(1) # Feature 2 & 3: Moving averages (trend indicators)
df['mavg_5'] = df['Close'].rolling(5).mean() df['mavg_20'] =
df['Close'].rolling(20).mean() # Feature 4: Volatility (risk regime indicator)
df['volatility'] = df['daily_return'].rolling(5).std() # Target: 1 = UP tomorrow, 0 =
DOWN tomorrow df['target'] = (df['daily_return'].shift(-1) > 0).astype(int) df =
df.dropna() # Remove rows with missing values
```

6.2 Statistical Analysis

```
returns = df['daily_return'] * 100 print(f'Mean: {returns.mean():.4f}%') # Average
daily return print(f'Median: {returns.median():.4f}%') # Middle value print(f'Std Dev:
{returns.std():.4f}%') # Daily volatility print(f'Skewness: {returns.skew():.4f}') #
Asymmetry of distribution print(f'Kurtosis: {returns.kurt():.4f}') # Fat-tail measure
```

6.3 Model Training & Evaluation

```
from sklearn.ensemble import RandomForestClassifier from sklearn.model_selection import
train_test_split from sklearn.metrics import accuracy_score, classification_report
features = ['prev_return', 'mavg_5', 'mavg_20', 'volatility'] X, y = df[features],
df['target'] # Split preserving time order (no shuffle!) X_train, X_test, y_train,
y_test = train_test_split( X, y, test_size=0.2, shuffle=False ) # Train: 100 trees vote
on each prediction model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train) # Evaluate on unseen test data predictions =
model.predict(X_test) print('Accuracy:', accuracy_score(y_test, predictions))
```

```
print(classification_report(y_test, predictions))
```

7. Planned Improvements

The current model uses four simple price-based features. The roadmap below outlines improvements planned to push accuracy toward 55-57%:

Priority	Improvement	Why It Helps
High	Add RSI (Relative Strength Index)	Detects overbought/oversold conditions
High	Add MACD indicator	Captures momentum shifts across timeframes
High	Add order book imbalance	Direct signal of buying vs selling pressure
Medium	Use 3-5 years of data	More examples for the model to learn from
Medium	Tune hyperparameters (GridSearchCV)	Optimizes tree depth, features per split
Medium	Try XGBoost or LightGBM	Often outperforms Random Forest on tabular data
Low	Implement LSTM neural network	Captures long-range temporal dependencies
Low	Add VIX and put/call ratio	Options market fear/greed signals
Low	Walk-forward validation	More realistic backtesting methodology

8. References

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This project was built as a learning exercise in quantitative finance and machine learning. Nothing in this report constitutes financial advice.